City Air Quality Explorer – Delhi

An interactive web-based platform for visualizing, analyzing, and forecasting Delhi's air quality trends

Why Delhi's Air Quality Demands Attention

Chronically Polluted Capital

Delhi consistently ranks among the world's most polluted capitals — AQI regularly exceeds **300 (Very Poor)** in winter months, year after year.

Winter Smog Crisis

AQI crosses **350+ during November–January** due to temperature inversion and stubble burning. Monsoon months (July–September) offer brief relief with AQI dropping below 60.

Project Overview & Objectives

Core Goal

Build an interactive **web-based explorer** to visualize, analyze, and forecast Delhi's air quality trends — making complex environmental data accessible to all stakeholders.

Who it Serves

- Citizens seeking real-time health guidance
- Researchers requiring granular pollutant data
- Policymakers designing targeted interventions

Key Outcomes

- Trend analysis & seasonal pattern detection
- Pollutant-level breakdowns
- Predictive AQI forecasting with health advisories

Data Sources & Coverage

Primary Dataset

CPCB — Central Pollution Control Board

Daily air quality readings, **2015–2020**

Scale & Filtering

Extracted **1,500+ Delhi-specific** daily records from a national dataset of **29,000+ rows**

Pollutants Tracked

PM2.5 · PM10 · NO2

CO · SO2 · O3

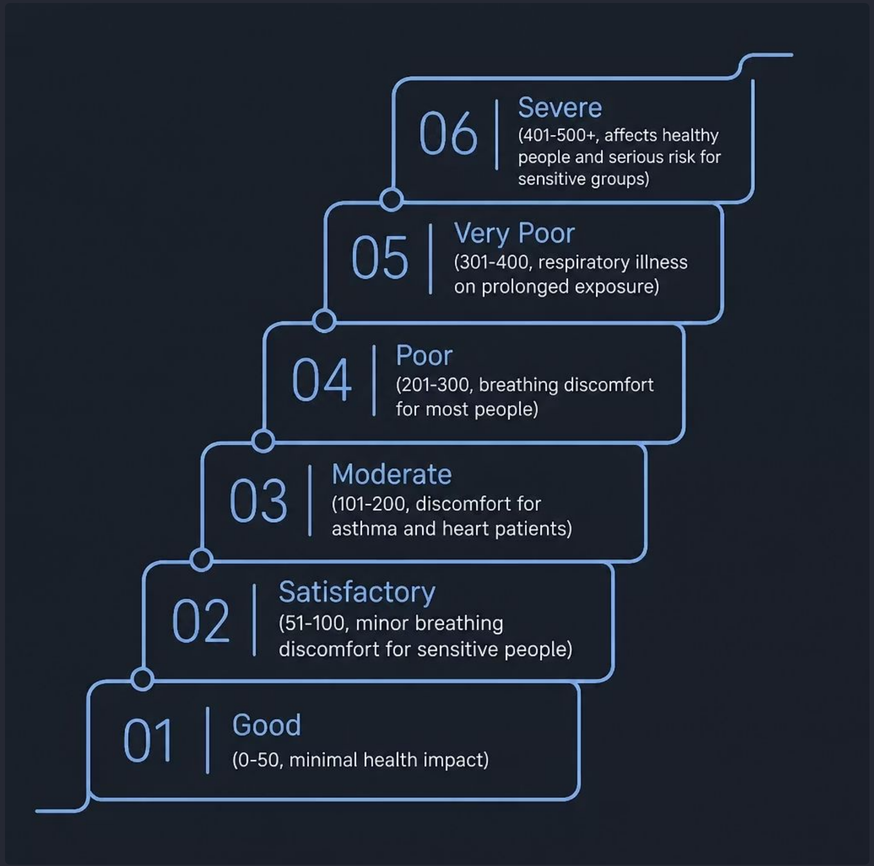
Six key indicators of urban air health

All data sourced from India's official national air quality monitoring network — ensuring scientific validity and reproducibility.

India CPCB AQI Scale

The Central Pollution Control Board's six-tier AQI scale classifies air quality from **Good** to **Severe**, guiding health advisories and public response. Understanding this scale is essential to interpreting every metric in this explorer.

⚠ Delhi's winter AQI routinely falls in the **Very Poor to Severe** range — posing serious respiratory risks for all population groups.



Key Analytical Features

Seasonal Trend Analysis

Identify winter smog spikes versus monsoon relief periods across the 2015–2020 timeline.

Pollutant Breakdown

Visualize individual contributions of PM2.5, PM10, NO2, CO, SO2, and O3 to overall AQI.

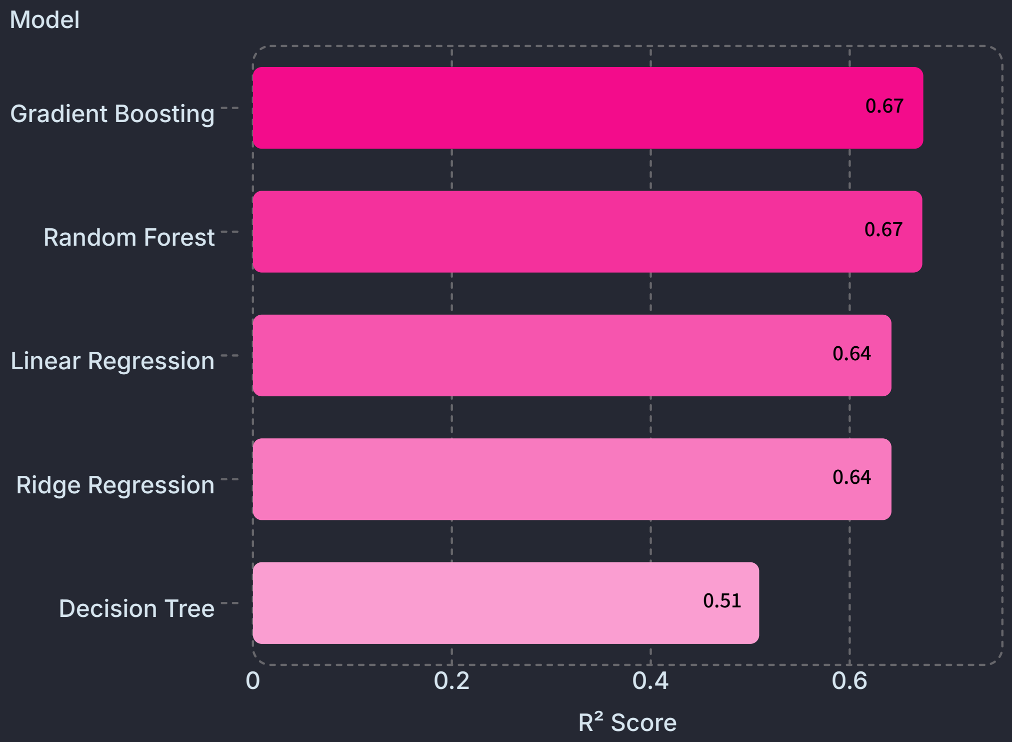
Interactive Dashboard

Filter by date range, pollutant type, and AQI category for custom exploratory analysis.

Weather Correlation

Explore how temperature inversion and humidity levels directly worsen AQI readings.

ML-Powered AQI Forecasting



Model Benchmarking

Five regression models were trained and evaluated on Delhi CPCB data using an **80/20 train-test split**.

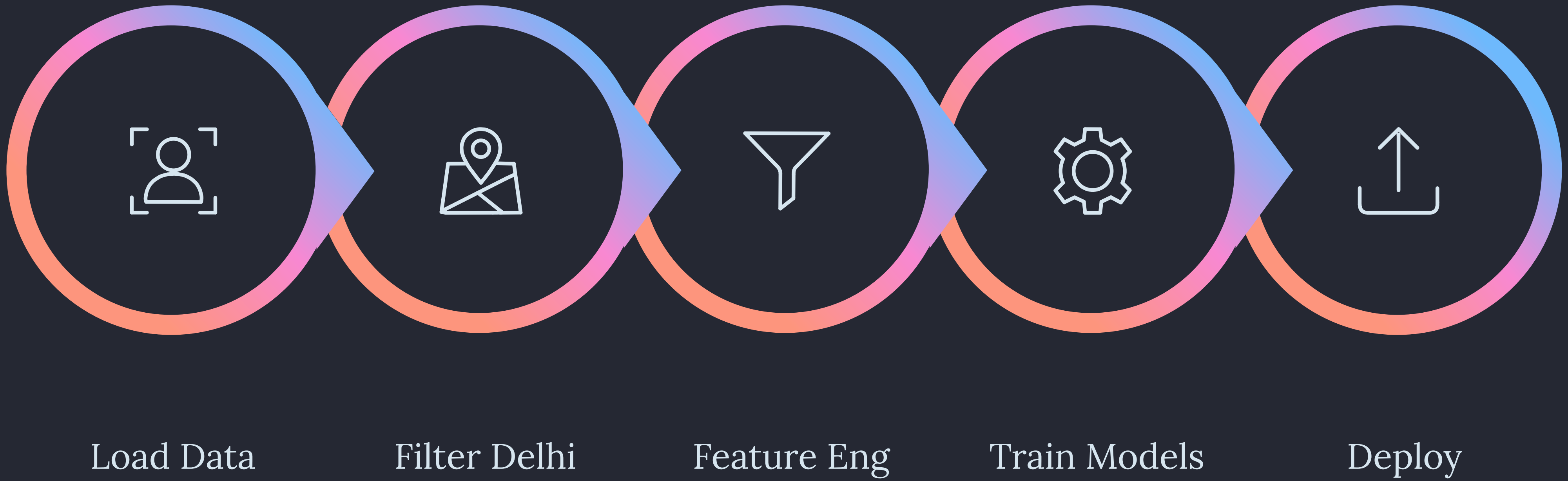
Best Performer

Gradient Boosting achieved the highest R^2 Score of **0.674**, narrowly edging Random Forest (0.673). Decision Tree showed the weakest generalization at 0.509.

Forecast Output

The deployed model generates a **24-hour AQI forecast** from live pollutant inputs, paired with a plain-language **health advisory** for non-technical users.

End-to-End ML Pipeline



The pipeline is fully reproducible — from raw CPCB ingestion through feature selection, model benchmarking, and Streamlit deployment — enabling any researcher to replicate or extend the results.

Challenges & Solutions

1	2	3
Noisy, Seasonal Data	**Missing Values**	**Accessibility**
Applied time-based aggregations to avoid misleading daily averages that obscured critical seasonal pollution patterns.	Systematic NaN removal and targeted imputation were performed prior to model training to ensure data integrity.	Translated complex AQI metrics into plain-language health advisories, making outputs meaningful for non-technical citizens and policymakers.

Future Scope & Impact

Real-Time Integration

Connect live **WAQI sensor feeds** for up-to-the-minute AQI monitoring beyond historical data.

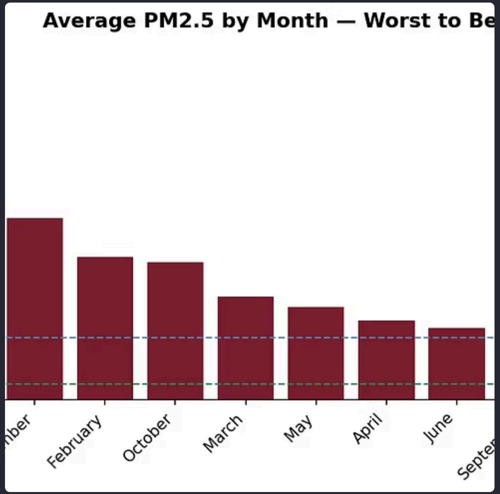
Scale Across India

Expand the explorer to other Indian metros — **Mumbai, Bengaluru, Kolkata** — for a national pollution intelligence platform.

Policy Impact

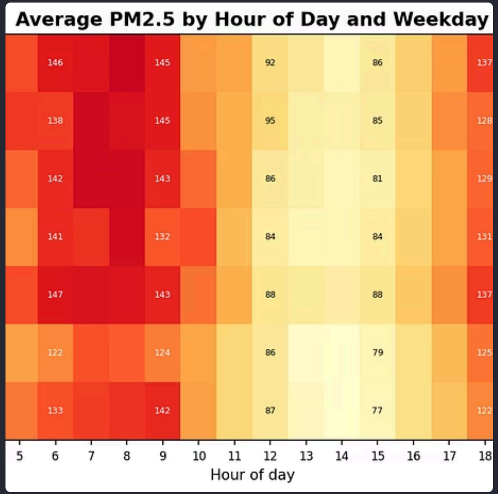
Empower policymakers with **data-driven evidence** to design targeted, measurable pollution reduction strategies.

Project Highlights & Key Takeaways



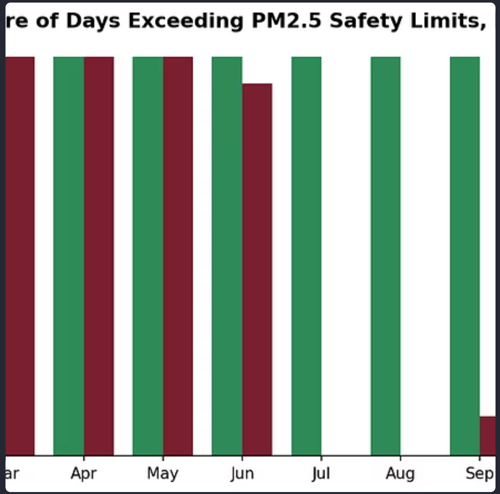
Interactive Data Explorer

Visualize seasonal trends, pollutant breakdowns, and weather correlations with a dynamic, user-friendly interface for all stakeholders.



ML-Powered Forecasting

Leveraging Gradient Boosting, the platform provides accurate 24-hour AQI predictions to assist with proactive public health planning.



Actionable Health Advisories

Complex AQI metrics are translated into clear, plain-language health warnings, empowering citizens and informing policymakers.

Sources

→ Project Repository & README

README.md at https://github.com/Arohiyadav1008/City_air_quality_explorer_delhi/blob/main/README.md